Performance Testing

Date	14 July 2026
Team ID	-
Project Name	Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Manual Testing, Streamlit, Python Performance Logs
Type of Testing	Functional Performance Testing
Target Module / API	Audio Upload, Speech-to-Text, Semantic Analysis, PDF Generation
Test Environment	Local Windows 11 System, Python 3.14, Streamlit
Test Date	Local Windows 11 System, Python 3.14, Streamlit

Step 2: Test Scenarios

S. No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Upload short audio (30 sec)	1	30 sec	Successful analysis and report generation

2	Upload medium audio (1 min)	1	60 sec	Correct transcription and similarity score
3	Upload unsupported audio format	1	10 sec	Error message displayed
4	Generate PDF Report	1	15 sec	PDF downloaded successfully

Step 3: Performance Test Results

S. No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.6 sec	Pass	Fast UI response
2	Response Time (Max)	< 5 seconds	3.8 sec	Pass	During Whisper processing
3	Throughput (Req/sec)	1	1.2 Request/sec	Pass	Stable performance
4	Error Rate	< 1%	0%	Pass	No runtime failures
5	CPU Utilization	< 80%	63%	Pass	Acceptable usage
6	Memory Utilization	< 80%	57%	Pass	Stable memory consumption

Step 4: Observations & Analysis

The performance testing results indicate that the Voice-Based Concept Understanding Analyser (VBCUA) operates efficiently under normal usage conditions. The application successfully uploaded audio files, generated accurate speech-to-text transcriptions, calculated semantic similarity scores, analysed audio characteristics, and produced downloadable PDF reports within acceptable response times. The only noticeable delay occurred during the initial loading of the Faster Whisper model and when processing longer audio recordings, which is expected due to the computational requirements of AI-based transcription. CPU and memory utilization remained within

acceptable limits, no critical errors were observed during testing, and the optimizations implemented, such as efficient model processing and automatic cleanup of temporary files, contributed to stable and reliable overall system performance.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):





